A RING OVER WHICH EVERY FAITHFULLY FLAT ETALE ALGEBRA HAS A RETRACTION HAS STRICTLY HENSELIAN LOCAL RINGS AT MAXIMAL IDEALS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file proetale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reduction to an isomorphism of the local rings through the description of the strict henselization, the separation of the primes over the maximal ideal, the construction of a faithfully flat etale map with a retraction via the openness of the image, and the closing flatness argument.

Editorial note. The label lemma-have-sections-strictly-henselian names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source proetale.tex –label lemma-have-sections-strictly-henselian –start 1660 –end 1700 –local-id 3580 returned RESULT=OK (L3580 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-proetale/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Description of the strict henselization of a local ring. Algebra, the lemma describing the strict henselization of a local ring as the limit over the etale neighbourhoods of a prime, used to reduce to an isomorphism of local rings. Supplies the reduction of the required statement to showing that the local map A_m -> B_q is an isomorphism.
- Quasi-finiteness of etale ring maps and the finiteness of the primes over a maximal ideal. Algebra, the lemma that an etale ring map is quasi-finite, and the elementary lemma allowing one to separate finitely many primes by an element. Supply the fact that the finitely many primes over m are maximal and admit no specializations, and the existence of an element separating them.
- Openness of the image of a flat finitely presented morphism. Algebra, the proposition that the image of a flat and finitely presented morphism of affine schemes is open. Supplies the openness of the image U of Spec(B) -> Spec(A), hence the element f with Spec(A) = U union D(f).
- Flat local homomorphisms of local rings are injective, and maps between etale algebras are flat. Algebra, the lemma that a flat local homomorphism of local rings is injective, and the lemma that a map between etale algebras

Date: stacks-project:proetale.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

over a ring is flat. Supply the conclusion that the retraction A_p -> B_q -> A_p forces the two local maps to be isomorphisms.

Source reference. Original lemma 00U5: sources/stacks-chow/tags.txt, source lines 1099–1099. Prerequisite pointer invoked at proof line(s) 1678; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00U7: sources/stacks-chow/tags.txt, source lines 1101–1101. Prerequisite pointer invoked at proof line(s) 1692; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 00DS: sources/stacks-chow/tags.txt, source lines 508–508. Prerequisite pointer invoked at proof line(s) 1681; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04GW: sources/stacks-chow/tags.txt, source lines 5847–5847. Prerequisite pointer invoked at proof line(s) 1672; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 00I1: sources/stacks-chow/tags.txt, source lines 663–663. Prerequisite pointer invoked at proof line(s) 1685; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let A be a ring such that every faithfully flat étale ring map $A \to B$ has a retraction. Then every local ring of A at a maximal ideal is strictly henselian.*

Proof. Let $\mathfrak{m}$ be a maximal ideal of A. Let $A \to B$ be an étale ring map and let $\mathfrak{q} \subset B$ be a prime lying over $\mathfrak{m}$. By the description of the strict henselization $A^{sh}_{\mathfrak{m}}$ in Algebra, Lemma 04GW it suffices to show that $A_{\mathfrak{m}} = B_{\mathfrak{q}}$. Note that there are finitely many primes $\mathfrak{q} = \mathfrak{q}_1, \mathfrak{q}_2, \ldots, \mathfrak{q}_n$ lying over $\mathfrak{m}$ and there are no specializations between them as an étale ring map is quasi-finite, see Algebra, Lemma 00U5. Thus $\mathfrak{q}_i$ is a maximal ideal and we can find $g \in \mathfrak{q}_2 \cap \ldots \cap \mathfrak{q}_n$, $g \notin \mathfrak{q}$ (Algebra, Lemma 00DS). After replacing B by B_g we see that $\mathfrak{q}$ is the only prime of B lying over $\mathfrak{m}$. The image $U \subset \mathrm{Spec}(A)$ of $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ is open (Algebra, Proposition 00I1). Thus the complement $\mathrm{Spec}(A) \setminus U$ is closed and we can find $f \in A$, $f \notin \mathfrak{p}$ such that $\mathrm{Spec}(A) = U \cup D(f)$. The ring map $A \to B \times A_f$ is faithfully flat and étale, hence has a retraction $\sigma : B \times A_f \to A$ by assumption on A. Observe that σ is étale, hence flat as a map between étale A-algebras (Algebra, Lemma 00U7). Since $\mathfrak{q}$ is the only prime of $B \times A_f$ lying over A we find that $A_{\mathfrak{p}} \to B_{\mathfrak{q}}$ has a retraction which is also flat. Thus $A_{\mathfrak{p}} \to B_{\mathfrak{q}} \to A_{\mathfrak{p}}$ are flat local ring maps whose composition is the identity. Since a flat local homomorphism of local rings is injective we conclude these maps are isomorphisms as desired. □